

Lecture Notes III

Course: Functional Analysis

TA: Morales C. Ronny

Topic: Measures

Theorem 1.23 (Properties of a measure).

i) If $A \subseteq B$, then $\mu(A) \leq \mu(B)$. If in addition $\mu(A) < \infty$, then

$$\mu(B \setminus A) = \mu(B) - \mu(A).$$

ii) $\mu(A) + \mu(B) = \mu(A \cup B) + \mu(A \cap B)$.

iii) (**σ -subadditivity**) $\mu\left(\bigcup_{n \in \mathbb{N}} A_n\right) \leq \sum_{n \in \mathbb{N}} \mu(A_n)$.

iv) (**Continuity from below and above**) If (A_n) is an increasing [respectively, decreasing] sequence in $\mathcal{M}$, then

$$\mu\left(\bigcup_{n \in \mathbb{N}} A_n\right) = \lim_{n \rightarrow \infty} \uparrow \mu(A_n), \quad \left[\text{respectively, } \mu\left(\bigcap_{n \in \mathbb{N}} A_n\right) = \lim_{n \rightarrow \infty} \downarrow \mu(A_n) \right].$$

Proof. i) Since $A \subseteq B$ we can write $B = A \cup (B \setminus A)$ so that

$$\mu(B) = \mu(A) + \mu(B \setminus A) \geq \mu(A)$$

and if $\mu(A) < \infty$, then we have $\mu(B) - \mu(A) = \mu(B \setminus A)$.

ii) Consider the decomposition $A \cup B = (A \setminus (A \cap B)) \cup ((B \setminus A \cap B)) \cup (A \cap B)$ and treat the cases $\mu(A \cap B) < \infty, \mu(A \cap B) = \infty$ separately.

iii) Suppose $(E_n) \subset \mathcal{M}$, then define

$$D_1 = \emptyset \quad \text{and} \quad D_k = \bigcup_{i=1}^{k-1} E_i$$

whence

$$D_n^\sim = \bigcap_{i=1}^{n-1} E_i^\sim$$

and thus $(E_n \setminus D_n)$ is a disjoint sequence of measurable sets whose union is

$$\bigcup_{n \in \mathbb{N}} (E_n \setminus D_n) = \bigcup_{n \in \mathbb{N}} (E_n \cap D_n^\sim) = \bigcup_{n \in \mathbb{N}} \left(E_n \cap \left(\bigcap_{i=1}^{n-1} E_i^\sim \right) \right) = \bigcup_{n \in \mathbb{N}} E_n,$$

whence

$$\mu\left(\bigcup_{n \in \mathbb{N}} E_n\right) = \mu\left(\bigcup_{n \in \mathbb{N}} (E_n \setminus D_n)\right) = \sum_{n \in \mathbb{N}} \mu(E_n \setminus D_n) \leq \sum_{n \in \mathbb{N}} \mu(E_n),$$

where the last inequality follows from i) and the fact that $E_n \setminus D_n \subset E_n$.

iv) Exercises.